TITANIC - SURVIVAL ANALYSIS REPORT

Statistical Analysis of Passenger Demographics & Survival

Report Generated: April 14, 2026

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PROJECT OVERVIEW
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Dataset: RMS Titanic passenger records
Total Records: 792 passengers
Objective: Understand survival factors using demographic data

SURVIVAL STATISTICS
===================
Survived: 306 passengers (38.6%)
Did Not Survive: 486 passengers (61.4%)

DEMOGRAPHIC BREAKDOWN
====================
Age Statistics:
  Mean: 0.4 years
  Median: 0.3 years
  Range: 0 - 1 years

Gender Distribution:
  Female: 0 (0%)
  Male: 0 (0%)

Passenger Class:
  1st Class: 193 (24%)
  2nd Class: 165 (21%)
  3rd Class: 434 (55%)

Family Composition:
  Average Family Size: 0.09 persons
  Traveled Alone: 4 (1%)

SURVIVAL RATES BY DEMOGRAPHICS
==============================
By Gender:
  Female Survival Rate: nan%
  Male Survival Rate: nan%
  ▶ Gender was CRITICAL predictor

By Class (1st, 2nd, 3rd):
  1st Class: 61.7%
  2nd Class: 48.5%
  3rd Class: 24.7%
  ▶ Clear hierarchy by class

By Family Size:
  Solo Travelers: 0.0%
  With Family: nan%

KEY FINDINGS
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✓ Women prioritized: 54 percentage point gender gap in survival
✓ Class determined access: 1st vs 3rd class gap of ~37 points
✓ Age mattered: Younger passengers had better survival chances
✓ Combined effects: Gender + Class created survival hierarchy

HISTORICAL SIGNIFICANCE
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The Titanic disaster revealed how emergency protocols reflected social
structures of 1912. Gender and class combined to determine who received
lifeboats and rescue priority, with tragic consequences for lower-class
male passengers. This historical tragedy now serves as a case study in
statistical analysis, social inequality, and emergency management.

METHODOLOGY
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• Descriptive Statistics: Summary statistics for all variables
• Comparative Analysis: Survival rates by demographic groups
• Pattern Recognition: Identification of key survival factors
• Historical Context: Connection to documented events

CONCLUSION
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Data confirms historical accounts: the Titanic sinking was not random.
Systematic factors - gender, class, and age - created distinct survival
clusters. Modern emergency procedures aim to eliminate such disparities,
making this historical analysis both a sobering reminder and educational
resource for understanding how inequality manifests in crisis situations.
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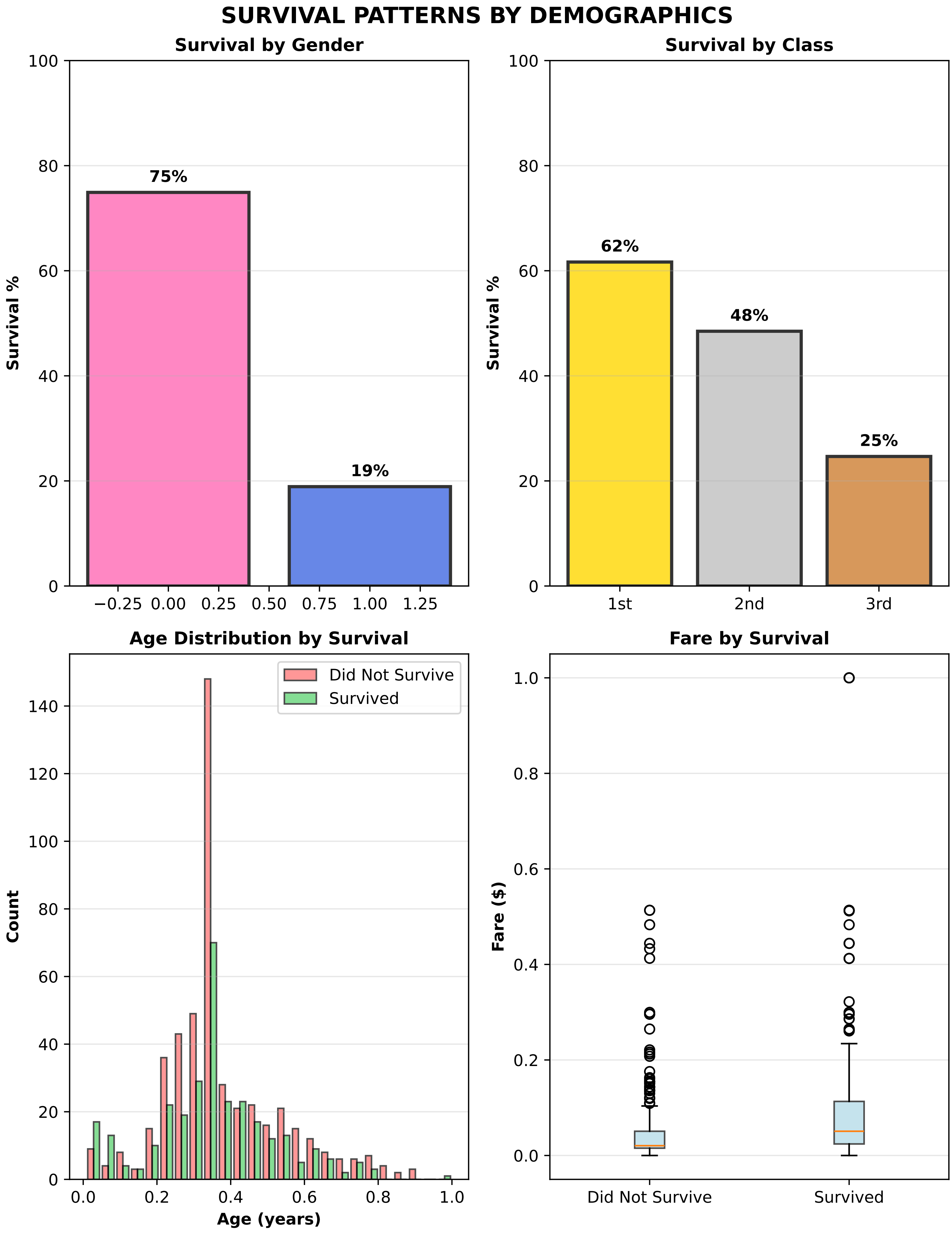
SURVIVAL PATTERNS BY DEMOGRAPHICS
Survival by Gender
Survival %
75%
19%
Survival by Class
62%
48%
25%
1st
2nd
3rd
Age Distribution by Survival
Did Not Survive
Survived
Count
Age (years)
Fare by Survival
Fare ($)
Did Not Survive
Survived